INCH-POUND

MIL-DTL-28840/28G
w/AMENDMENT 1
8 February 2019
SUPERSEDING
MIL-DTL-28840/28G
21 November 2013

DETAIL SPECIFICATION SHEET

CONNECTORS, ELECTRICAL, CIRCULAR, THREADED, HIGH DENSITY, HIGH SHOCK, SHIPBOARD, CRIMP CONTACTS, PLUG ASSEMBLY, 90º, CLASSES DJ, DJS, PJ, TJ, AND ZJ

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-28840.

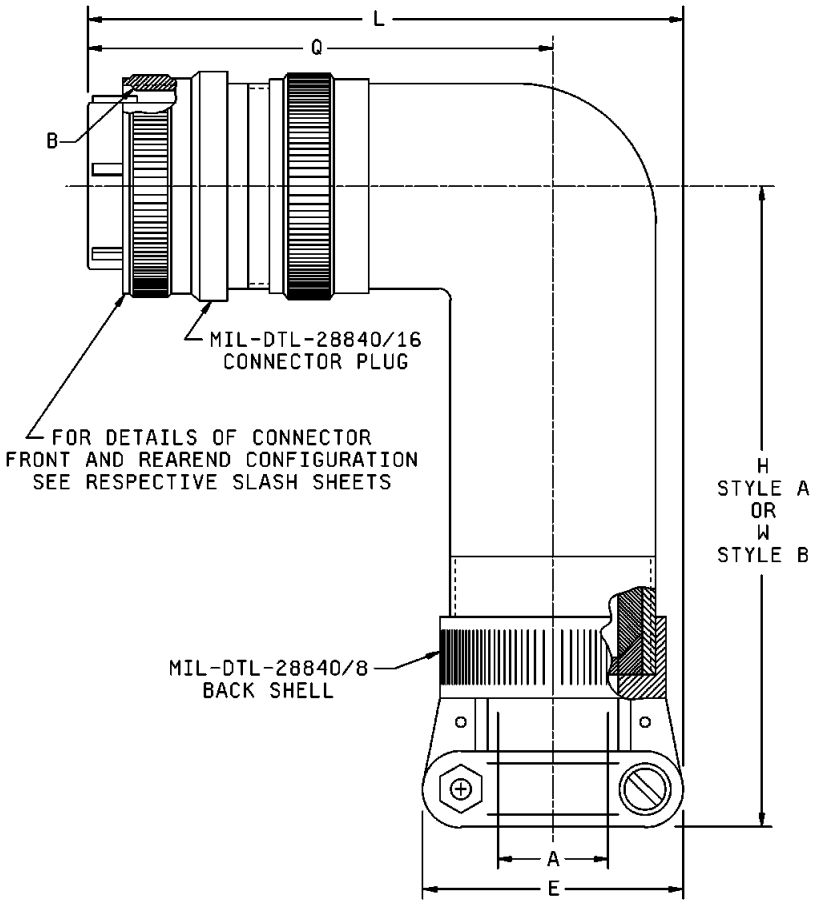


FIGURE 1. Dimensions and configuration.

AMSC N/A

FSC 5935

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Designator (see note 3)	Assembly Part or Identifying Number (PIN) (see note 6)	Connector plug (PIN) (see note 6)	Backshell assembly (PIN) (see note 6)	A Cable entry		B thread class 2B	E ref
				max	min		
A (11)	M28840/28*A*AA**	M28840/16*A***	M28840/801*A	.312 (7.92)	.188 (4.77)	.750-.1P -.2L-D.S.	.938 (23.82)
A (11)	M28840/28*A*AB**	M28840/16*A***	M28840/801*B				
A (11)	M28840/28*A*BA**	M28840/16*A***	M28840/824*A	.438 (11.13)	.281 (7.14)	.750-.1P -.2L-D.S.	1.125 (28.57)
A (11)	M28840/28*A*BB**	M28840/16*A***	M28840/824*B				
A (11)	M28840/28*A*CA**	M28840/16*A***	M28840/825*A	.562 (14.27)	.344 (8.74)	.750-.1P -.2L-D.S.	1.312 (33.32)
A (11)	M28840/28*A*CB**	M28840/16*A***	M28840/825*B				
B (13)	M28840/28*B*AA**	M28840/16*B***	M28840/802*A	.312 (7.92)	.188 (4.77)	.875-.1P -.2L-D.S.	.938 (23.82)
B (13)	M28840/28*B*AB**	M28840/16*B***	M28840/802*B				
B (13)	M28840/28*B*BA**	M28840/16*B***	M28840/803*A	.438 (11.13)	.281 (7.14)	.875-.1P -.2L-D.S.	1.125 (28.57)
B (13)	M28840/28*B*BB**	M28840/16*B***	M28840/803*B				
B (13)	M28840/28*B*CA**	M28840/16*B***	M28840/826*A	.562 (14.27)	.344 (8.74)	.875-.1P -.2L-D.S.	1.312 (33.32)
B (13)	M28840/28*B*CB**	M28840/16*B***	M28840/826*B				
B (13)	M28840/28*B*DA**	M28840/16*B***	M28840/827*A	.625 (15.87)	.375 (9.52)	.875-.1P -.2L-D.S.	1.312 (33.32)
B (13)	M28840/28*B*DB**	M28840/16*B***	M28840/827*B				
C (15)	M28840/28*C*AA**	M28840/16*C***	M28840/805*A	.438 (11.13)	.281 (7.14)	1.062-.1P -.2L-D.S.	1.125 (28.57)
C (15)	M28840/28*C*AB**	M28840/16*C***	M28840/805*B				
C (15)	M28840/28*C*BA**	M28840/16*C***	M28840/806*A	.562 (14.27)	.344 (8.74)	1.062-.1P -.2L-D.S.	1.312 (33.32)
C (15)	M28840/28*C*BB**	M28840/16*C***	M28840/806*B				
C (15)	M28840/28*C*CA**	M28840/16*C***	M28840/828*A	.625 (15.87)	.375 (9.52)	1.062-.1P -.2L-D.S.	1.312 (33.32)
C (15)	M28840/28*C*CB**	M28840/16*C***	M28840/828*B				
C (15)	M28840/28*C*DA**	M28840/16*C***	M28840/807*A	.750 (19.05)	.438 (11.12)	1.062-.1P -.2L-D.S.	1.593 (40.46)
C (15)	M28840/28*C*DB**	M28840/16*C***	M28840/807*B				
C (15)	M28840/28*C*EA**	M28840/16*C***	M28840/804*A	.312 (7.92)	.188 (4.77)	1.062-.1P -.2L-D.S.	.938 (23.82)
C (15)	M28840/28*C*EB**	M28840/16*C***	M28840/804*B				

FIGURE 1. Dimensions and configuration – Continued.

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Designator (see note 3)	Assembly (PIN) (see note 6)	Connector plug (PIN) (see note 6)	Backshell assembly (PIN) (see note 6)	H max style A	N max style B	L max	Q max
A (11)	M28840/28*A*AA**	M28840/16*A***	M28840/801*A	5.332 (135.43)	--	4.139 (105.13)	3.306 (83.97)
A (11)	M28840/28*A*AB**	M28840/16*A***	M28840/801*B	--	2.962 (75.23)		
A (11)	M28840/28*A*BA**	M28840/16*A***	M28840/824*A	5.332 (135.43)	--		3.306 (83.97)
A (11)	M28840/28*A*BB**	M28840/16*A***	M28840/824*B	--	2.962 (75.23)		
A (11)	M28840/28*A*CA**	M28840/16*A***	M28840/825*A	5.457 (138.61)	--	4.389 (111.48)	3.431 (87.15)
A (11)	M28840/28*A*CB**	M28840/16*A***	M28840/825*B	--	3.087 (78.41)		
B (13)	M28840/28*B*AA**	M28840/16*B***	M28840/802*A	5.332 (135.43)	--	4.139 (105.13)	3.306 (83.97)
B (13)	M28840/28*B*AB**	M28840/16*B***	M28840/802*B	--	2.962 (75.23)		
B (13)	M28840/28*B*BA**	M28840/16*B***	M28840/803*A	5.332 (135.43)	--	4.139 (105.13)	3.306 (83.97)
B (13)	M28840/28*B*BB**	M28840/16*B***	M28840/803*B	--	2.962 (75.23)		
B (13)	M28840/28*B*CA**	M28840/16*B***	M28840/826*A	5.457 (138.61)	--	4.389 (111.48)	3.431 (87.15)
B (13)	M28840/28*B*CB**	M28840/16*B***	M28840/826*B	--	3.087 (78.41)		
B (13)	M28840/28*B*DA**	M28840/16*B***	M28840/827*A	5.682 (144.32)	--	4.639 (117.83)	3.556 (90.32)
B (13)	M28840/28*B*DB**	M28840/16*B***	M28840/827*B	--	3.312 (84.12)		
C (15)	M28840/28*C*AA**	M28840/16*C***	M28840/805*A	5.457 (138.61)	--	4.389 (111.48)	3.431 (87.15)
C (15)	M28840/28*C*AB**	M28840/16*C***	M28840/805*B	--	3.087 (78.41)		
C (15)	M28840/28*C*BA**	M28840/16*C***	M28840/806*A	5.457 (138.61)	--		3.431 (87.15)
C (15)	M28840/28*C*BB**	M28840/16*C***	M28840/806*B	--	3.087 (78.41)		
C (15)	M28840/28*C*CA**	M28840/16*C***	M28840/828*A	5.682 (144.32)	--	4.639 (117.83)	3.556 (90.32)
C (15)	M28840/28*C*CB**	M28840/16*C***	M28840/828*B	--	3.312 (84.12)		
C (15)	M28840/28*C*DA**	M28840/16*C***	M28840/807*A	5.713 (145.11)	--		3.556 (90.32)
C (15)	M28840/28*C*DB**	M28840/16*C***	M28840/807*B	--	3.343 (84.91)		
C (15)	M28840/28*C*EA**	M28840/16*C***	M28840/804*A	5.457 (138.61)	--	4.389 (111.48)	3.431 (87.15)
C (15)	M28840/28*C*EB**	M28840/16*C***	M28840/804*B	--	3.087 (78.41)		

FIGURE 1. Dimensions and configuration – Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T19:45Z
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Designator (see note 3)	Assembly (PIN) (see note 6)	Connector plug (PIN) (see note 6)	Backshell assembly (PIN) (see note 6)	A Cable entry		B thread class 2B	E ref
				max	min		
D (17)	M28840/28*D*AA**	M28840/16*D***	M28840/808*A	.625 (15.87)	.375 (9.52)	1.125-.1P -.2L-D.S.	1.312 (33.32)
D (17)	M28840/28*D*AB**	M28840/16*D***	M28840/808*B				
D (17)	M28840/28*D*BA**	M28840/16*D***	M28840/809*A	.750 (19.05)	.438 (11.12)	1.125-.1P -.2L-D.S.	1.593 (40.46)
D (17)	M28840/28*D*BB**	M28840/16*D***	M28840/809*B				
D (17)	M28840/28*D*CA**	M28840/16*D***	M28840/810*A	.938 (23.82)	.625 (15.87)	1.125-.1P -.2L-D.S.	1.750 (44.45)
D (17)	M28840/28*D*CB**	M28840/16*D***	M28840/810*B				
E (19)	M28840/28*E*AA**	M28840/16*E***	M28840/811*A	.625 (15.87)	.375 (9.52)	1.312-.1P -.2L-D.S.	1.312 (33.32)
E (19)	M28840/28*E*AB**	M28840/16*E***	M28840/811*B				
E (19)	M28840/28*E*BA**	M28840/16*E***	M28840/812*A	.938 (23.82)	.625 (15.87)	1.312-.1P -.2L-D.S.	1.750 (44.45)
E (19)	M28840/28*E*BB**	M28840/16*E***	M28840/812*B				
F (23)	M28840/28*F*AA**	M28840/16*F***	M28840/813*A	.750 (19.05)	.438 (11.12)	1.500-.1P -.2L-D.S.	1.593 (40.46)
F (23)	M28840/28*F*AB**	M28840/16*F***	M28840/813*B				
F (23)	M28840/28*F*BA**	M28840/16*F***	M28840/814*A	.938 (23.82)	.625 (15.87)	1.500-.1P -.2L-D.S.	1.750 (44.45)
F (23)	M28840/28*F*BB**	M28840/16*F***	M28840/814*B				
F (23)	M28840/28*F*CA**	M28840/16*F***	M28840/815*A	1.188 (30.18)	.812 (20.62)	1.500-.1P -.2L-D.S.	2.093 (53.16)
F (23)	M28840/28*F*CB**	M28840/16*F***	M28840/815*B				

FIGURE 1. Dimensions and configuration – Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T19:45Z
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Designator (see note 3)	Assembly (PIN) (see note 6)	Connector plug (PIN) (see note 6)	Backshell assembly (PIN) (see note 6)	H max style A	N max style B	L max	Q max
D (17)	M28840/28*D*AA**	M28840/16*D***	M28840/808*A	5.682 (144.32)	--	4.639 (117.83)	3.556 (90.32)
D (17)	M28840/28*D*AB**	M28840/16*D***	M28840/808*B	--	3.312 (84.12)		
D (17)	M28840/28*D*BA**	M28840/16*D***	M28840/809*A	5.713 (145.11)	--		3.556 (90.32)
D (17)	M28840/28*D*BB**	M28840/16*D***	M28840/809*B	--	3.343 (84.91)		
D (17)	M28840/28*D*CA**	M28840/16*D***	M28840/810*A	5.807 (147.50)	--		3.556 (90.32)
D (17)	M28840/28*D*CB**	M28840/16*D***	M28840/810*B	--	3.437 (87.30)		
E (19)	M28840/28*E*AA**	M28840/16*E***	M28840/811*A	5.682 (144.32)	--		3.556 (90.32)
E (19)	M28840/28*E*AB**	M28840/16*E***	M28840/811*B	--	3.312 (84.12)		
E (19)	M28840/28*E*BA**	M28840/16*E***	M28840/812*A	5.832 (148.13)	--	4.889 (124.18)	3.681 (93.50)
E(19)	M28840/28*E*BB**	M28840/16*E***	M28840/812*B	--	3.462 (87.93)		
F (23)	M28840/28*F*AA**	M28840/16*F***	M28840/813*A	5.738 (145.75)	--		3.681 (93.50)
F (23)	M28840/28*F*AB**	M28840/16*F***	M28840/813*B	--	3.368 (85.55)		
F (23)	M28840/28*F*BA**	M28840/16*F***	M28840/814*A	5.832 (148.13)	--		3.681 (93.50)
F (23)	M28840/28*F*BB**	M28840/16*F***	M28840/814*B	--	3.462 (87.93)		
F (23)	M28840/28*F*CA**	M28840/16*F***	M28840/815*A	6.019 (152.88)	--		3.681 (93.50)
F (23)	M28840/28*F*CB**	M28840/16*F***	M28840/815*B	--	3.649 (92.68)		

FIGURE 1. Dimensions and configuration – Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T19:45Z
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Designator (see note 3)	Assembly number (PIN) (see note 6)	Connector plug (PIN) (see note 6)	Backshell assembly (PIN) (see note 6)	A Cable entry		B thread class 2B	E ref
				max	min		
G (25)	M28840/28*G*AA**	M28840/16*G***	M28840/816*A	.938 (23.82)	.625 (15.87)	1.625-.1P -.2L-D.S.	1.750 (44.45)
G (25)	M28840/28*G*AB**	M28840/16*G***	M28840/816*B				
G (25)	M28840/28*G*BA**	M28840/16*G***	M28840/817*A	1.188 (30.18)	.812 (20.62)	1.625-.1P -.2L-D.S.	2.093 (53.16)
G (25)	M28840/28*G*BB**	M28840/16*G***	M28840/817*B				
G (25)	M28840/28*G*CA**	M28840/16*G***	M28840/818*A	1.250 (31.75)	.875 (22.22)	1.625-.1P -.2L-D.S.	2.093 (53.16)
G (25)	M28840/28*G*CB**	M28840/16*G***	M28840/818*B				
G (25)	M28840/28*G*DA**	M28840/16*G***	M28840/819*A	1.380 (35.05)	1.000 (25.40)	1.625-.1P -.2L-D.S.	2.343 (59.51)
G (25)	M28840/28*G*DB**	M28840/16*G***	M28840/819*B				
H (29)	M28840/28*H*AA**	M28840/16*H***	M28840/820*A	1.188 (30.18)	.812 (20.62)	1.812-.1P -.2L-D.S.	2.093 (53.16)
H (29)	M28840/28*H*AB**	M28840/16*H***	M28840/820*B				
H (29)	M28840/28*H*BA**	M28840/16*H***	M28840/821*A	1.380 (35.05)	1.000 (25.40)	1.812-.1P -.2L-D.S.	2.343 (59.51)
H (29)	M28840/28*H*BB**	M28840/16*H***	M28840/821*B				
J (33)	M28840/28*J*AA**	M28840/16*J***	M28840/822*A	1.380 (35.05)	1.000 (25.40)	2.000-.1P -.2L-D.S.	2.343 (59.51)
J (33)	M28840/28*J*AB**	M28840/16*J***	M28840/822*B				
J (33)	M28840/28*J*BA**	M28840/16*J***	M28840/823*A	1.625 (41.27)	1.250 (31.75)	2.000-.1P -.2L-D.S.	2.750 (69.85)
J (33)	M28840/28*J*BB**	M28840/16*J***	M28840/823*B				

FIGURE 1. Dimensions and configuration – Continued.

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Designator (see note 3)	Assembly (PIN) (see note 6)	Connector plug (PIN) (see note 6)	Backshell assembly (PIN) (see note 6)	H max style A	N max style B	L max	Q max
G (25)	M28840/28*G*AA**	M28840/16*G***	M28840/816*A	5.832 (148.13)	--	4.889 (124.18)	3.681 (93.50)
G (25)	M28840/28*G*AB**	M28840/16*G***	M28840/816*B	--	3.462 (87.93)		
G (25)	M28840/28*G*BA**	M28840/16*G***	M28840/817*A	6.019 (152.88)	--		3.681 (93.50)
G (25)	M28840/28*G*BB**	M28840/16*G***	M28840/817*B	--	3.649 (92.68)		
G (25)	M28840/28*G*CA**	M28840/16*G***	M28840/818*A	6.144 (156.06)	--	5.139 (130.53)	3.806 (96.67)
G (25)	M28840/28*G*CB**	M28840/16*G***	M28840/818*B	--	3.774 (95.86)		
G (25)	M28840/28*G*DA**	M28840/16*G***	M28840/819*A	6.176 (156.87)	--		3.806 (96.67)
G (25)	M28840/28*G*DB**	M28840/16*G***	M28840/819*B	--	3.806 (96.67)		
H (29)	M28840/28*H*AA**	M28840/16*H***	M28840/820*A	6.144 (156.06)	--		3.806 (96.67)
H (29)	M28840/28*H*AB**	M28840/16*H***	M28840/820*B	--	3.774 (95.86)		
H (29)	M28840/28*H*BA**	M28840/16*H***	M28840/821*A	6.176 (156.87)	--		3.806 (96.67)
H (29)	M28840/28*H*BB**	M28840/16*H***	M28840/821*B	--	3.806 (96.67)		
J (33)	M28840/28*J*AA**	M28840/16*J***	M28840/822*A	6.426 (163.22)	--	5.639 (143.23)	4.056 (103.02)
J (33)	M28840/28*J*AB**	M28840/16*J***	M28840/822*B	--	4.056 (103.02)		
J (33)	M28840/28*J*BA**	M28840/16*J***	M28840/823*A	6.701 (170.21)	--		4.056 (103.02)
J (33)	M28840/28*J*BB**	M28840/16*J***	M28840/823*B	--	4.331 (110.01)		

NOTES:

1. Dimensions are in inches.
2. Metric equivalents are given for information only.
3. Shell sizes are provided within parentheses for information and are not a part of the designator.
4. Dimensions apply after plating.
5. Normal mating key position shown. Alternate key positions in accordance with MIL-DTL-28840.
6. Assembly shall be made of a connector and backshell of the same material.

FIGURE 1. Dimensions and configuration – Continued.

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REQUIREMENTS:

Dimensions and configuration: See figure 1.

Classes: DJ, DJS, PJ, TJ, and ZJ. Corresponding class codes: C, E, G, M, and U, respectively.

Insert arrangements: See the figure of MIL-STD-1698 corresponding to the desired shell size.

Contacts: Pin or socket, crimp type, styles P, S, D, E, F or G, in accordance with MIL-DTL-28840.

Shell polarization: 1 thru 6, key position.

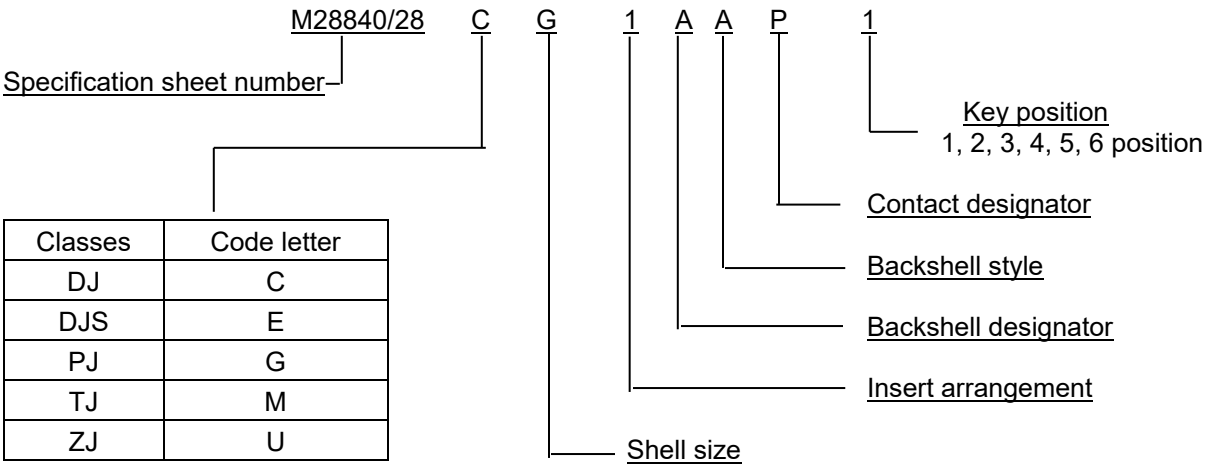
Marking: Part or Identifying Number (PIN) on shell.

Mating counterpart: Mating connectors in accordance with MIL-DTL-28840/10, MIL-DTL-28840/11, MIL-DTL-28840/12, MIL-DTL-28840/14, MIL-DTL-28840/20 and MIL-DTL-28840/21.

Installation and removal tool: In accordance SAE-AS81969/33 and SAE-AS81969/34.

PIN's shown indicate aluminum alloy hardware. If corrosion resistant steel parts are required, the connector and accessory part numbers must be changed to designate corrosion resistant hardware.

PIN example:



Classes	Code letter
DJ	C
DJS	E
PJ	G
TJ	M
ZJ	U

RESTRICTIONS:

Classes DJ and DJS are inactive for NAVSEA new design, unless approved by NAVSEA.

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Amendment notations. The margins of this specification are marked with vertical lines to indicate modifications generated by this amendment. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations.

Referenced documents. In addition to MIL-DTL-28840, this document references the following:

MIL-DTL-28840/8
MIL-DTL-28840/10
MIL-DTL-28840/11
MIL-DTL-28840/12
MIL-DTL-28840/14
MIL-DTL-28840/16
MIL-DTL-28840/20
MIL-DTL-28840/21
SAE-AS81969/33
SAE-AS81969/34
MIL-STD-1698

CONCLUDING MATERIAL

Custodians:
Army - AT
Navy - EC
Air Force - 85
DLA - CC

Preparing activity:
DLA - CC

(Project 5935-2019-008)

Review activities:
Army – AV, MI

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

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